DSA0405 - Fundamentals of Data Science

Total points 18/25

CO2-AT3 Assessment Tool MCQs

1. A dataset follows a normal distribution with μ = 500 and σ = 50. What is the probability that a randomly chosen value is above 600? 1/1

a) 2.28%

b) 5%

c) 16%

d) 0.13%

2.A dataset contains salaries that follow a normal distribution with μ = 75,000 and σ = 10,000. 1/1

What is the Z-score for a salary of 95,000?

a) 1.5

b) 2.0

c) 2.5

d) 3.0

3.A Hadoop cluster stores 500 GB of data with a block size of 128 MB and a replication factor of 3. 1/1

How much total storage is required in the cluster?

a) 500 GB

b) 1000 GB

c) 1500 GB

d) 2000 GB

4.A dataset contains 1 million records, and 20% of them belong to Class A. 0/1

If we randomly sample 5000 records, what is the expected number of Class A records in the sample?

a) 1000

b) 1250

c) 1500

d) 2000

Correct answer

b) 1250

5. A MapReduce job processes 1 TB of data using 100 nodes. If each node processes 10 GB per hour, how long will the entire job take? 0/1

a) 1 hour

b) 2 hours

c) 10 hours

d) 20 hours

Correct answer

b) 2 hours

6.A dataset has 1000 samples, where:

0/1

600 belong to Class A 400 belong to Class B

A split results in two child nodes:
Left: 300 Class A, 50 Class B Right: 300 Class A, 350 Class B

What is the information gain?

a) 0.125

b) 0.278

c) 0.541

d) 0.672

Correct answer

c) 0.541

7. A dataset has 10 features. PCA retains 95% variance with the top 4 components. What percentage of variance do the remaining components contribute?

1/1

a) 2%

b) 5%

c) 10%

d) 20%

8. A dataset contains values ranging from 1 to 1,000,000. Which transformed value does 10,000 map to if we apply a log transformation using base 10?

1/1

a) 2

b) 3

c) 5

d) 4

9.A dataset follows a normal distribution with μ = 150 and σ = 20. What percentage of values lie beyond 3 standard deviations from the mean (both tails combined)?

1/1

a) 0.3%

b) 1.35%

c) 2.5%

d) 5%

10.A dataset has 1 million records and 5% of them contain at least one missing value. If we remove all rows with missing values, how many records remain?

0/1

a) 900,000

b) 950,000

c) 990,000

d) 975,000

Correct answer

d) 975,000

11. A classification model gives Precision = 0.80 and Recall = 0.60. What is the F1-score?

1/1

a) 0.65

b) 0.70

c) 0.75

d) 0.80

12. Consider the following NumPy matrix:

1/1

A=[2 4 3 6]

a) 0

b) 2

c) 6

d) 12

13.A Pandas DataFrame with 1 million rows is sorted using

1/1

df.sort_values(by='column1'). The sorting algorithm used by Pandas is:

a) Merge Sort

b) Quick Sort

c) Tim Sort

d) Heap Sort

14. A Matplotlib plot renders 1 million data points using plt.scatter(). The

1/1

plot is slow. What is the best way to optimize performance?

a) Use plt.plot() instead of plt.scatter()

b) Enable marker="." in plt.scatter()

c) Use ax.hexbin() instead of plt.scatter()

d) Increase figure size

15. A Pandas DataFrame has 10 million rows and 100 columns. Which method is fastest for selecting specific rows based on conditions?

1/1

a) df[df['column1'] > 100]

b) df.loc[df['column1'] > 100]

c) df.query('column1 > 100')

d) df.apply(lambda x: x['column1'] > 100, axis=1)

16. A NumPy array of shape (100,000, 10) is stored using float64 data type. What is the total memory consumption in MB?

1/1

a) 8 MB

b) 80 MB

c) 160 MB

d) 320 MB

17. A K-Means clustering model is run on 1 million points with K=5. The convergence time is 10 minutes. If K is doubled to 10, assuming all other factors remain the same, what is the expected increase in execution time?

1/1

a) No change

b) 2x increase

c) 4x increase

d) 10x increase

18.A NumPy array of shape (200,000, 5) is stored using float32 instead of float64. How much memory is saved in MB? 0/1

a) 2 MB

b) 4 MB

c) 8 MB

d) 16 MB

Correct answer

c) 8 MB

19. A dataset contains 100,000 data points, and a scatter plot is generated using Matplotlib. Each marker takes 100 bytes of memory. What is the total memory usage for the scatter plot? 1/1

a) 1 MB

b) 5 MB

c) 10 MB

d) 100 MB

20. A Pandas DataFrame with 1 million rows is sorted based on a single column. The sorting algorithm used by Pandas is Timsort (O(n log n)). If sorting takes 5 seconds, how long would it take for 10 million rows? 0/1

a) 10 seconds

b) 50 seconds

c) 25 seconds

d) 60 seconds

Correct answer

c) 25 seconds

21. A Pandas DataFrame contains 20,000 rows. After applying a filter that removes 40% of the data, how many rows remain? 1/1

A) 10,000

B) 12,000

C) 14,000

D) 16,000

22. A NumPy array contains numbers from 1 to 500. What is the mean of this dataset? 1/1

- A) 245
- B) 250
- C) 255
- D) 275

23. A dataset contains 5000 values distributed into 50 bins in a histogram. What is the average number of values per bin? 1/1

A) 80

B) 90

C) 100

D) 110

24.Which sub-package of SciPy is specifically designed for handling vector quantization and K-Means clustering algorithms? 1/1

A) scipy.optimize

B) scipy.spatial

C) scipy.cluster

D) scipy.linalg

25.What core data science task are both NumPy and SciPy explicitly built to solve together? 0/1

A) Web scraping and API deployment.

B) Mathematical and numerical analysis.

C) Static data visualization and dashboarding.

D) Database management and query optimization.

Correct answer

D) Database management and query optimization.